

UNIT I: Introduction to NLP and Text Representation

1. What is a corpus in NLP, and what are the different types of corpora?
2. Identify and explain the key components of Natural Language Processing.
3. How are regular expressions used in NLP tasks?
4. Define corpora and discuss their importance in NLP applications.
5. Explain the process and techniques used in word tokenization.
6. Differentiate between stemming and lemmatization, providing use-case examples.
7. Describe the purpose and methods of sentence segmentation.
8. Define minimum edit distance and its application in NLP.
9. How does the OneHotEncoder represent textual data numerically?
10. Explain the Bag of Words model and its limitations.
11. What are n-grams? Describe bigrams and trigrams with examples.
12. Define TF-IDF and explain how it improves upon Bag of Words.
13. What is Word2Vec and how does it capture semantic information?
14. Compare and contrast count-based vs. prediction-based text representation methods.
15. Discuss challenges in preprocessing textual data for NLP tasks.

UNIT II: Naive Bayes, Text Classification, and Sentiment Analysis

1. Explain the Naive Bayes classifier and its assumptions.
2. Describe the steps involved in training a Naive Bayes classifier.
3. How is Naive Bayes optimized specifically for sentiment analysis?

4. Discuss the use of Naive Bayes as a language model.
5. What are precision, recall, and F-measure? Why are they important in evaluating classifiers?
6. How does the independence assumption impact Naive Bayes classification?
7. What role does smoothing play in Naive Bayes training?
8. Explain how Naive Bayes handles text classification for multiple classes.
9. Compare Naive Bayes with other classification algorithms in NLP.
10. What are common challenges faced when applying Naive Bayes to sentiment analysis?
11. How is text data prepared and represented for Naive Bayes classification?
12. Define false positive and false negative errors in the context of classifier evaluation.
13. Discuss methods to handle imbalanced datasets in sentiment classification.
14. Explain the significance of model interpretability in Naive Bayes classifiers.
15. What are the advantages and disadvantages of using Naive Bayes for NLP tasks?

UNIT III: Hidden Markov Models, Speech Recognition, and Part-of-Speech Tagging

1. Outline the architecture of a typical speech recognition system.
2. What is a Hidden Markov Model (HMM) and how is it applied in speech recognition?
3. Explain the Viterbi algorithm and its role in decoding HMM sequences.
4. Describe advanced methods used for decoding in speech recognition.
5. How is acoustic processing performed on speech signals?
6. Discuss how acoustic probabilities are computed in HMM-based speech systems.
7. What are the steps involved in training a speech recognizer?
8. Describe techniques used for waveform generation in speech synthesis.

9. Explain the various word classes in English and their syntactic roles.
10. What is Part-of-Speech (POS) tagging, and why is it important?
11. Compare rule-based, stochastic, and transformation-based POS tagging methods.
12. What are tagsets in POS tagging? Give examples of common tagsets used for English.
13. Discuss the challenges in POS tagging for ambiguous words.
14. How do POS tags contribute to higher-level NLP tasks?
15. Explain the significance of sentence segmentation in speech and text processing.

UNIT IV: Transformers and Large Language Models

1. What is a self-attention network and how does it function within Transformers?
2. Describe the multihead attention mechanism and its advantages.
3. Break down the components of a Transformer block.
4. Explain the concept of residual streams in Transformer architectures.
5. What is the purpose of the language modelling head in Transformer-based LLMs?
6. How do Transformers overcome limitations of traditional sequential models like RNNs?
7. Discuss how positional encoding is integrated into Transformer models.
8. What are the training objectives commonly used for large language models?
9. Explain how large language models use transformers for natural language understanding.
10. What are some challenges faced when scaling Transformer-based language models?
11. Discuss the role of attention weights in interpreting Transformer model outputs.
12. How do residual connections aid in deep Transformer training?

13. Define transfer learning and its importance in deploying large pretrained language models.
14. Describe a few applications of large language models in real-world NLP systems.
15. What future trends are anticipated in Transformer and LLM research?
appropriate for postgraduate courses on NLP, speech recognition, and language models.